

Diagnosis and management of acute hyperkalaemia

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Series explanation: State of the Art Reviews are commissioned on the basis of their relevance to academics and specialists in the US and internationally.

ABSTRACT

Hyperkalaemia is a common and potentially life threatening electrolyte abnormality because of its association with malignant cardiac arrhythmias and sudden cardiac arrest. This condition is frequently encountered across acute and chronic care settings and disproportionately affects patients with comorbid conditions like chronic kidney disease, heart failure, diabetes mellitus, and those receiving drugs that impair renal potassium excretion. The most frequent causes of hyperkalaemia include impaired renal potassium elimination, from chronic kidney disease or pharmacological inhibition of the renin-angiotensin-aldosterone system, and transcellular potassium shifts resulting from metabolic acidosis, tissue breakdown, insulin deficiency, or hypertonic states. Despite its prevalence and clinical importance, substantial variability exists in the approach to diagnosis and management of hyperkalaemia, reflecting differences in definitions, risk stratification, and therapeutic approaches across clinical settings. Management strategies have evolved in recent years with improved understanding of potassium physiology and the availability of newer potassium binding agents, necessitating an updated synthesis of the evidence. In this state of the art review, we summarise contemporary data on the epidemiology and pathophysiology of hyperkalaemia, identify key risk factors and diagnostic considerations, and review evidence based approaches to acute and chronic management, with particular attention to special populations and high risk clinical scenarios.

Introduction

Hyperkalaemia is a common and potentially life threatening electrolyte abnormality because of its association with arrhythmias and sudden cardiac arrest.¹ This condition is defined as an abnormally raised serum potassium concentration, typically ≥ 5.0 or 5.5 mmol/L in adults. However, toxicity may occur at lower thresholds because risk increases along a continuum rather than a discrete threshold, depending on the rate of increase and patient specific factors.²⁻⁴ The most frequent causes of hyperkalaemia include impaired renal potassium excretion (eg, chronic kidney disease (CKD), use of renin-angiotensin-aldosterone system (RAAS) inhibitors), transcellular shifts (eg, from acidosis, tissue breakdown, or insulin deficiency), and less commonly, increased potassium intake.⁵ Despite its prevalence and clinical importance, the diagnosis and management of hyperkalaemia remain varied across practice settings, and therapeutic strategies have continued to evolve with the emergence of newer drug treatment options.^{1,4}

In this state of the art review, we synthesise the current evidence on hyperkalaemia, with a focus on epidemiology, pathophysiology, risk factors,

diagnostic considerations, acute management strategies, and special populations. This review is intended for clinicians across primary care, emergency, and inpatient settings who regularly encounter patients with hyperkalaemia.

Source and selection of criteria

We searched PubMed, Ovid Medline, and the Cochrane Database of Systematic Reviews for articles published from inception to 1 December 2025 to identify studies on the diagnosis and management of hyperkalaemia. Search terms included “hyperkalemia” or “hyperkalaemia,” “hyperpotassemia” or “hyperpotassemias,” “high potassium” or “elevated potassium,” and “potassium abnormality.” We evaluated retrospective and prospective studies, randomised controlled trials, systematic reviews and meta-analyses, and guidelines. When available, systematic reviews and meta-analyses were preferentially selected, followed sequentially by randomised controlled trials, prospective studies, and retrospective studies. We excluded non-peer reviewed publications (including preprint articles), case series, case reports, and editorials. If articles pertained to similar topics, we selected more recent publications.

Table 1 | Definition of hyperkalaemia^{1 4-7}

Definition of hyperkalaemia*	European Resuscitation Council, UK Kidney Association, and Kidney Disease: Improving Global Outcomes	American Heart Association and American Academy of Family Physicians
Overall definition	Serum potassium ≥ 5.5 mmol/L	Serum potassium ≥ 5.0 mmol/L based on serum concentration and ECG findings
Mild	5.5-5.9 mmol/L	5.0-5.9 mmol/L without ECG changes
Moderate	6.0-6.4 mmol/L	5.0-5.9 mmol/L with ECG changes or 6.0-6.4 mmol/L without ECG changes
Severe	≥ 6.5 mmol/L	6.0-6.4 mmol/L with ECG changes or >6.5 mmol/L without ECG changes

ECG=electrocardiogram.

*American guidelines use mEq/L, which have been converted to mmol/L throughout (for potassium, 1 mmol/L=1 mEq/L).

Definition

There is no universally accepted definition of hyperkalaemia. The European Resuscitation Council, UK Kidney Association, and Kidney Disease: Improving Global Outcomes (KDIGO) definition is a serum potassium ≥ 5.5 mmol/L.^{4 6 7} The American Heart Association and the American Academy of Family Physicians use a serum potassium cut-off level of ≥ 5 mmol/L.^{1 5} Table 1 gives definitions based on severity. However, regardless of the definition, risk increases along a continuum rather than a discrete threshold.

Epidemiology

The prevalence of hyperkalaemia in the general population is estimated to be between 1.5% and 3%.⁸⁻¹⁰ Large United States claims and Medicare database studies report an annual prevalence of 1.5-2.7%, with higher rates (6.4-9.3%) in those experiencing CKD or heart failure, or both conditions.^{9 11} Prevalence is likely underestimated and depends on a variety of factors, including the definition of hyperkalaemia and laboratory sensitivities. Advanced age, CKD, heart failure, diabetes, hypertension, and a range of drugs are all associated with a higher risk and severity of hyperkalaemia.^{9 12 13}

Hyperkalaemia is frequently encountered in the emergency department and inpatient care, reflecting its strong association with acute illness, drug use, and underlying chronic disease.^{1 4} A cross-sectional claims database study found that, in 2011, approximately 814 000 visits to the emergency department were related to hyperkalaemia in the US.¹⁴ Reported inpatient rates of hyperkalaemia ranged from 1.1% to 10%, depending on the study population.^{15 16} Although data in the UK are limited, some studies have shown similar findings.^{16 17} This high prevalence highlights the importance of timely recognition, appropriate risk stratification, and standardised management pathways to prevent adverse outcomes.

Physiology and pathophysiology

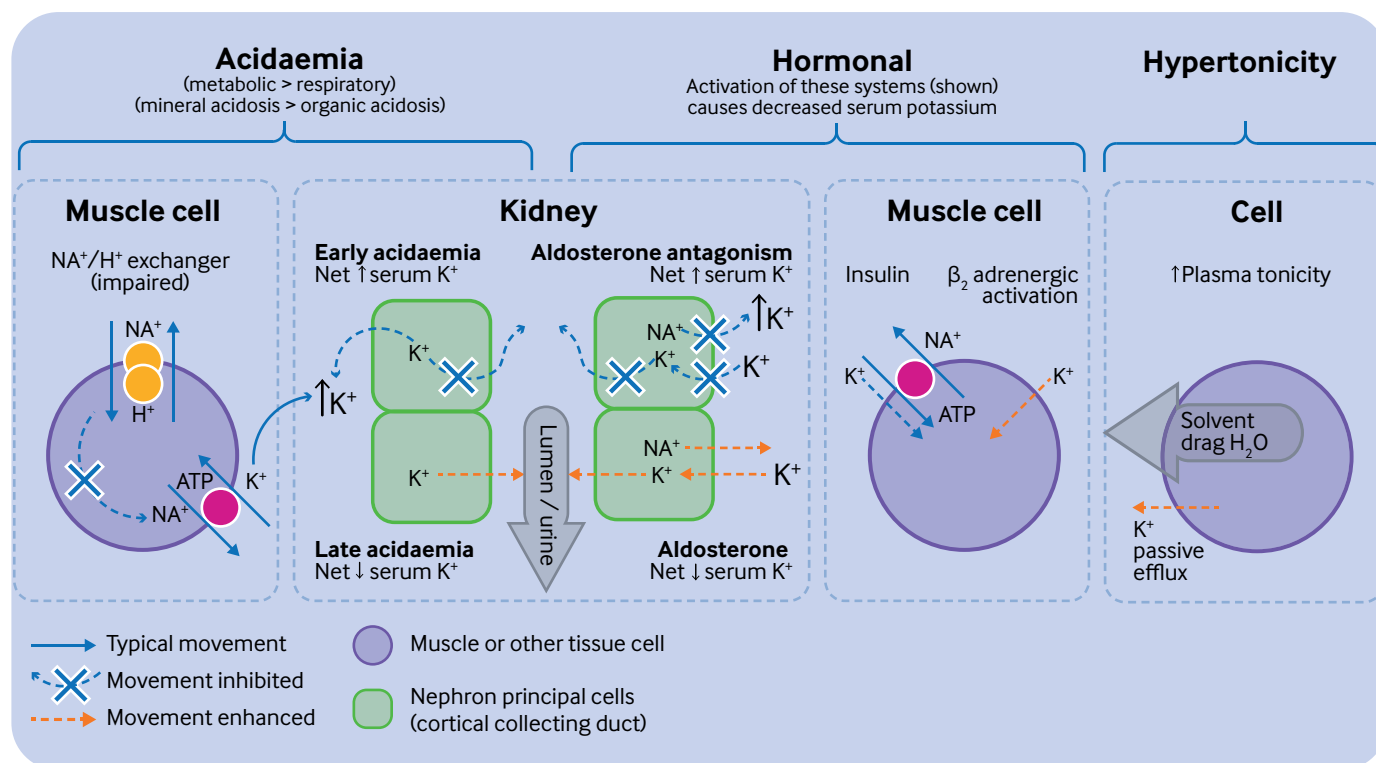
Potassium serves myriad roles in the body, but primarily regulates excitability of heart and other muscle tissues. Potassium is mostly intracellular (98%), and extracellular potassium (2%) is tightly controlled by homeostasis. Normal serum potassium

levels are widely accepted to lie between 3.5 and 5.0 mmol/L.^{1 4 7} Small deviations can have dramatic consequences as the therapeutic window is narrow, most notably fatal cardiac dysrhythmia and cardiac arrest.¹⁸ Consequently, extracellular and serum potassium is primarily controlled by an interplay of three mechanisms: acid-base balance, renal transport and excretion, and hormone signalling (fig 1).¹⁸

Acid-base balance plays a well established role in potassium homeostasis through its effects on transcellular potassium shifts and renal potassium handling.¹⁹ Potassium distribution is influenced by pH sensitive transporters.²⁰ In acidaemia, impaired Na^+/H^+ exchange reduces intracellular sodium, suppressing Na^+/K^+ -ATPase activity and promoting an extracellular potassium shift.²¹ Metabolic acidosis is thought to cause a greater degree of hyperkalaemia than respiratory acidosis, but the association is variable. Organic metabolic acidotic states (eg, diabetic ketoacidosis, lactic acidosis) may not produce marked hyperkalaemia despite considerable acidaemia because insulin deficiency, catecholamine effects, and renal potassium losses modify net potassium balance.^{19 21 22}

The kidneys also have an important role in potassium homeostasis. In addition to transcellular shifts, acid-base status exerts complex and time dependent effects on renal potassium handling. For example, early phases of metabolic acidosis lead to reduced urinary potassium excretion, while later stages of metabolic acidosis promote urinary potassium excretion and development of hypokalaemia.²¹ When serum potassium is raised, aldosterone is generated, increasing the expression of epithelial sodium channels and Na^+/K^+ -ATPase pumps in the distal nephron. The net effect of aldosterone is increased renal potassium secretion.²³ However, in conditions such as CKD and decompensated heart failure, reduced nephron function and impaired renal perfusion limit potassium secretion and predispose people to hyperkalaemia.²⁴

Several hormonal mechanisms are involved in potassium balance. Feedforward control describes hormone signalling in the postprandial phase. Insulin is the most well known factor, which in addition to facilitating glucose uptake, induces Na^+/K^+ -ATPase pump production, thereby shifting potassium intracellularly.^{18 21 24} Catecholamines,

Fig 1 | Mechanism of potassium homeostasis¹⁸

through β_2 adrenergic activation, also shift potassium intracellularly to compensate for potassium release during muscle contraction or sympathetic tone. In contrast, α adrenergic stimulation causes a release of potassium extracellularly but is typically negligible.²⁴ Additional hormone signalling pathways occur between the mesentery and the kidney in an anticipatory fashion to buffer potassium levels after a meal.²³ Despite this complex hormonal interplay, between 5% and 10% of potassium is secreted in the colon and excreted through the stool.²⁵ Therefore, considerable potassium losses can occur in severe diarrhoea.²⁶

A less commonly emphasised, but clinically important, mechanism contributing to hyperkalaemia is increased plasma tonicity. Hypertonic states promote transcellular water efflux, leading to extracellular potassium movement through solvent drag and concentration dependent passive efflux.²⁷ As cells lose water and undergo shrinkage, intracellular potassium concentration increases, further augmenting potassium efflux down its concentration gradient and contributing to increases in serum potassium levels. Clinically, this is most evident in hyperglycaemic crises (eg, diabetic ketoacidosis and hyperosmolar hyperglycaemic states) and, to a lesser extent, with the use of hypertonic agents such as mannitol, where osmotic water shifts may raise serum potassium independent of total body potassium stores.

In patients with kidney disease, the risk of hyperkalaemia is strongly correlated with declining

estimated glomerular filtration rate. The CKD Prognosis Consortium (2018) analysed more than one million patients across several international cohorts and showed increased hyperkalaemia risk with declining estimated glomerular filtration rate.²⁸ Although the association is non-linear, previous observational studies suggest that for each 15 mL/min/1.73 m² reduction in estimated glomerular filtration rate, the risk of hyperkalaemia (serum potassium >5.5 mmol/L) approximately doubles, with the highest risk observed at levels <15 mL/min/1.73 m².¹⁸ Although adaptive increases in distal nephron potassium secretion may partially compensate in earlier stages of CKD, these mechanisms become insufficient as kidney function deteriorates, particularly in the presence of concurrent illness, volume depletion, or potassium raising drugs.

Consequences of hyperkalaemia

Hyperkalaemia is associated with a range of negative clinical effects, including cardiac dysrhythmias and arrest, neuromuscular dysfunction, renal tubular acidosis, and increased mortality.²⁹⁻³² Hyperkalaemia associated hospital admissions are also extremely costly, accounting for approximately \$1.2bn (£0.9bn; €1.05bn) in inpatient costs in the US in 2014.³³ While there are limited epidemiological studies outside of the US, a Spanish retrospective cohort study of 1181 patients with heart failure and reduced ejection fraction experienced 438 episodes of hyperkalaemia in 262 patients, amounting to a cost of €89 178.82 for healthcare use over 64 months.³⁴

The most well established and concerning complication of hyperkalaemia is the development of cardiac dysrhythmias. A wide variety of conduction abnormalities may occur, including atrial and ventricular tachyarrhythmias and bradyarrhythmias. The exact potassium level at which dysrhythmias occur varies among patients.^{1 6 18 35 36} Cardiac conduction abnormalities arise because raised extracellular potassium reduces the resting membrane potential of cardiac myocytes, leading to partial depolarisation of the cell membrane. The inactivation of fast sodium channels slows phase 0 depolarisation, resulting in decreased conduction velocity through the atria, atrioventricular node, and ventricles.³⁷ As potassium levels rise, this manifests on the electrocardiogram (ECG) as peaked T waves, PR interval prolongation, QRS complex widening, and eventually a sine wave pattern, reflecting progressive conduction delay and risk of ventricular arrhythmias and asystole.^{37 38}

Studies show a strong association between hyperkalaemia and mortality in hospital.^{39 40} One large analysis of more than 900 000 inpatients showed all cause mortality increased markedly with each 0.1 mmol/L increase in low and high potassium (<4 and >5.0 mmol/L, respectively), in a U-shaped fashion.⁴¹ A meta-analysis of 27 cohort studies (n=1 217 986) found a similar U-shaped association, with the lowest risk of adverse effects at 4-4.5 mmol/L.²⁸ Compared with a reference level of 4.2 mmol/L, the adjusted hazard ratio for all cause mortality was 1.22 (95% confidence interval (CI) 1.15 to 1.29) at 5.5 mmol/L. A large retrospective cohort study of nearly 50 000 patients in hospital showed a significant risk of mortality when the index serum potassium was >6.0 mmol/L (adjusted odds ratio 5.0, 95% CI 2.9 to 8.6).⁴² Finally, a systematic review and meta-analysis including five studies (n=246 612) found a similar U-shaped curve in length of stay, with progressively longer stays observed at higher serum potassium concentrations. Compared with the reference group (3.5-5.0 mmol/L), the pooled raw differences of median lengths of stay were as follows: 4.45 (95% CI 2.71 to 6.91), 1.99 (0.03 to 3.94), 0.98 (0.91 to 1.05), 1.51 (1.03 to 2.0), 1 (0.75 to 1.25), and 2.76 (1.24 to 4.29) for patients with potassium levels of <2.5, 2.5 to <3.0, 3.0 to <3.5, <5 to 5.5, <5.5 to 6, and >6.0 mmol/L, respectively.⁴³ These associations persist even when hyperkalaemia occurs as a secondary finding, underscoring its role as a marker of illness severity and a potentially modifiable contributor to adverse outcomes in acute care settings. Additionally, hyperkalaemia is associated with long term adverse outcomes, with studies showing that during an index hospital visit, regardless of hospital admission, hyperkalaemia is associated with more frequent laboratory monitoring, outpatient visits, and need for dialysis, leading to increased healthcare system use and costs.⁴⁴⁻⁴⁶

Causes and risk factors for hyperkalaemia

Common risk factors for the development of acute hyperkalaemia can be categorised into a

few overarching themes: reduced renal excretion, transcellular shifts, and excessive intake. In general, worsening renal function, a higher baseline potassium level, and the presence of comorbidities like heart failure, diabetes mellitus, acute kidney injury, and CKD are associated with the development of hyperkalaemia.^{47 48}

Reduced renal excretion

Because 90% of potassium is eliminated through the kidneys, when acute kidney injury arises, so does hyperkalaemia.^{18 27} When the RAAS system is impaired because of underlying disease or pharmacological inhibition, the kidney's capacity to secrete potassium is diminished. The risk of hyperkalaemia is frequently compounded by the use of evidence based pharmacotherapies that are integral to reducing morbidity and mortality in conditions such as diabetes, hypertension, heart failure, and CKD.⁴⁸⁻⁵² These agents include angiotensin converting enzyme inhibitors, angiotensin receptor blockers, mineralocorticoid receptor antagonists, and direct renin inhibitors, all of which decrease aldosterone mediated potassium secretion in the distal nephron.⁵ Additional drugs commonly implicated in hyperkalaemia include potassium sparing diuretics (eg, amiloride, triamterene), non-steroidal anti-inflammatory drugs, heparin, trimethoprim, calcineurin inhibitors, succinylcholine, and beta adrenergic blockers, which interfere with renal potassium handling or transcellular potassium shifts through diverse mechanisms.⁵³ Acute kidney injury and CKD decrease glomerular filtration, while hypoaldosteronism impairs potassium secretion within the distal collecting duct of the nephron.^{47 49} Several drugs further reduce renal excretion. Drugs like RAAS inhibitors, potassium sparing diuretics, non-steroidal anti-inflammatory agents, and others.

Transcellular shifts

Transcellular shifts of potassium contribute to hyperkalaemia in the setting of acidosis, insulin deficiency (eg, diabetic ketoacidosis), tissue breakdown (eg, rhabdomyolysis, trauma or crush injury, tumour lysis syndrome), and certain drugs (eg, digoxin, succinylcholine).^{49 54-56} Additionally, hyperkalaemia has been documented in the setting of electrical burns and combat associated injuries.^{57 58}

Increased potassium intake

Increased intake alone is infrequently associated with the development of hyperkalaemia because of the capacity of the kidney to excrete potassium.^{49 59} However, in the setting of CKD or acute kidney injury, the kidneys' capacity to excrete potassium may be diminished. Iatrogenic hyperkalaemia can occur, often secondary to potassium supplementation (oral or intravenous) or blood transfusions.^{4 49} Potassium-containing salt substitutes and herbal products (eg, alfalfa, dandelion, milkweed) may also contribute to exogenous potassium exposure.^{7 49}

Special populations warrant heightened vigilance. Patients receiving haemodialysis are at particularly high risk, influenced by dietary intake, dialysis frequency and duration, glycaemic control, and constipation.⁶⁰⁻⁶² Polypharmacy and several comorbid conditions further amplify risk, especially when several potassium raising drugs are used concurrently (eg, angiotensin converting enzyme inhibitors, aldosterone antagonists, non-steroidal anti-inflammatory drugs, beta blockers).^{44 48 63-66}

Pseudohyperkalaemia

Finally, pseudohyperkalaemia represents a falsely raised serum potassium without true physiological abnormality.⁶⁷ The most frequent causes are prolonged blood storage time before laboratory analysis or poor storage.⁶⁸ Additionally, fist clenching, a haemolysed sample (eg, tourniquet time exceeding one minute, fine gauge needles, mechanical trauma), thrombocytosis, and leucocytosis can result in pseudohyperkalaemia.^{59 68} Potassium-containing fluids may also contaminate samples if drawn through the same line as the infusion. Figure 2 presents a summary of the causes, risk factors, and special considerations that should trigger healthcare professionals to have a higher suspicion for the presence of hyperkalaemia.

Diagnosis of acute hyperkalaemia

Symptoms

Most patients with hyperkalaemia do not have symptoms, with the abnormality identified on screening ECG or routine laboratory analysis.⁵ Paraesthesia and muscular fasciculations may be early signs of hyperkalaemia.⁴⁹ Paralysis, arrhythmias, and cardiac arrest may be lethal presentations.^{27 39} Less common symptoms include nausea and fatigue. Usually, hyperkalaemia muscle weakness is ascending, starting in the legs and progressing to the trunk, sometimes resembling Guillain-Barré syndrome.⁴⁹ A retrospective cohort study of 923 patients presenting with severe hyperkalaemia found that 46.8% had symptoms attributable to hyperkalaemia, with 20.3% experiencing cardiac arrest and 35.2% presenting with dysrhythmias.³⁹ In some patients, these symptoms may be more attributable to the underlying disease process rather than hyperkalaemia itself, such as muscle cramping in rhabdomyolysis.

ECG findings

ECG is an important and rapid tool in diagnosing and risk stratifying hyperkalaemia; however, it is not a reliable predictor of outcome.⁶⁹ Furthermore, it has a low sensitivity for detecting hyperkalaemia and does not reliably correlate with serum potassium concentrations.^{69 70} These abnormalities may be a

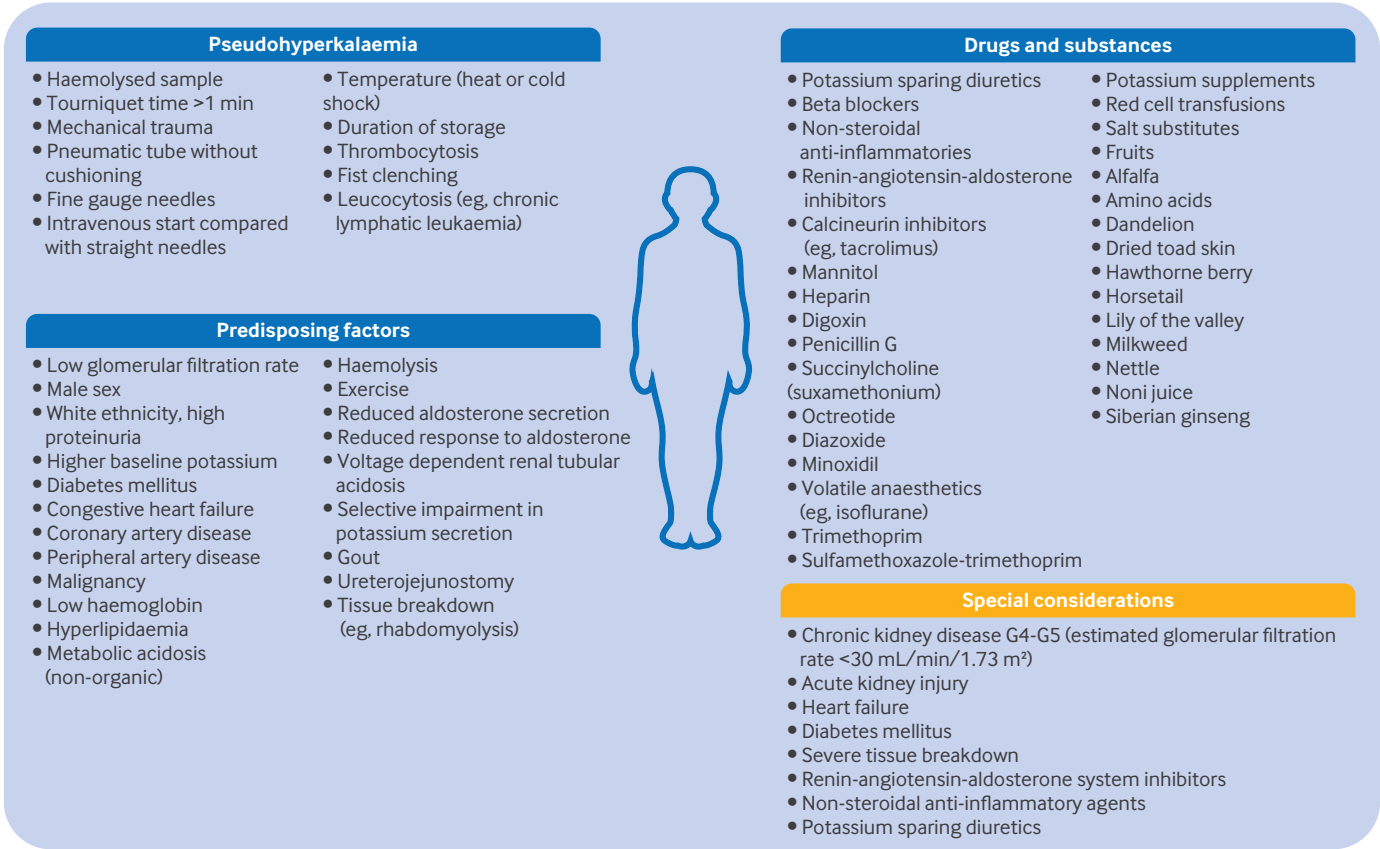


Fig 2 | Causes, risk factors, and special considerations for hyperkalaemia

marker of rapid increases in serum potassium rather than a specific raised potassium level. Therefore, the threshold to start treatment should not be based on ECG findings alone and treatment decisions should not be delayed while awaiting classic ECG changes to manifest.⁵ One of the earliest ECG signs of raised potassium is a “peaked” T wave (fig 3). At higher potassium levels, the ECG may progress to PR prolongation and QRS widening. A common and serious presentation of severe hyperkalaemia (eg, potassium >6.5 mmol/L, ECG changes, or both) is junctional bradycardia and a sine wave pattern. This rhythm may be mistaken for a primary cardiac cause and typically resolves promptly with calcium administration. In one retrospective analysis of 188 patients with severe hyperkalaemia, prolonged QRS interval (relative risk 4.7, 95% CI 2.0 to 11.6), bradycardia (12.3, 6.7 to 22.6), and

junctional rhythm (7.5, 5.3 to 11.1) were the most likely to be associated with a serious adverse event such as ventricular fibrillation, cardiopulmonary resuscitation, or death within six hours.⁷¹ Furthermore, a prospective, observational study of 203 patients found that peaked T waves occurred in <25% of patients with a potassium of 5.5-7.0 mmol/L and in only 35% of patients with potassium >7.0 mmol/L, while widened QRS occurred in <13% of patients with a potassium level of 5.5-7.0 mmol/L.¹⁵ High risk hyperkalaemia is generally defined as a serum potassium concentration >6.5 mmol/L or pathological ECG changes, or both. However, risk stratification should also consider the rate of potassium increase, underlying comorbidities, and clinical manifestations such as muscle weakness, paralysis, or haemodynamic instability.^{4 5 49}

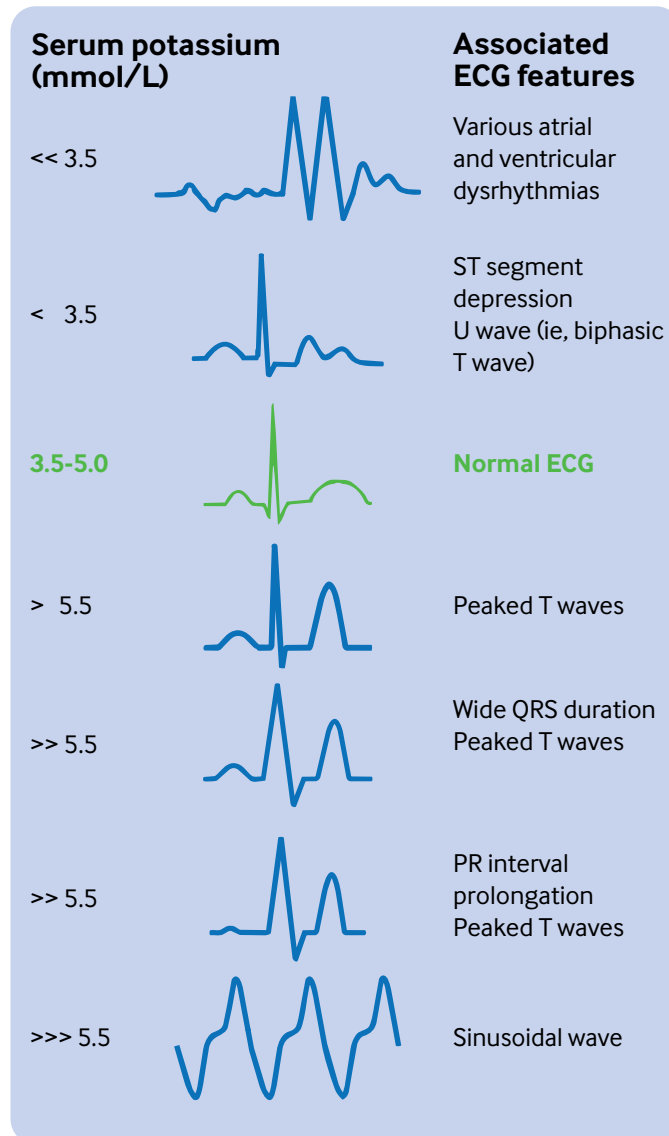


Fig 3 | Changes in electrocardiogram (ECG) across potassium concentrations

Serum testing

There are several options for serum potassium testing. Plasma and whole blood potassium measurements differ markedly in sample processing and susceptibility to preanalytical error in diagnosing acute hyperkalaemia.⁷² Plasma potassium is measured from anticoagulated blood that has been centrifuged to remove cellular components, limiting potassium release from cells, and providing a reliable estimate of circulating potassium.⁷³ Whole blood potassium is measured without centrifugation, most commonly using point-of-care

analysers, and includes intact cellular elements. Although whole blood testing provides rapid results, it is more susceptible to handling related variability and does not detect haemolysis, which may lead to pseudohyperkalaemia.^{74 75} In most patients, whole blood potassium values are slightly lower than plasma, though clinically meaningful discrepancies can occur owing to pseudohyperkalaemia.^{74 75} Point-of-care analysers (measure whole blood) can be used in the laboratory or at the bedside and consist of a machine and specific cartridges that measure potassium. These devices are most useful to obtain

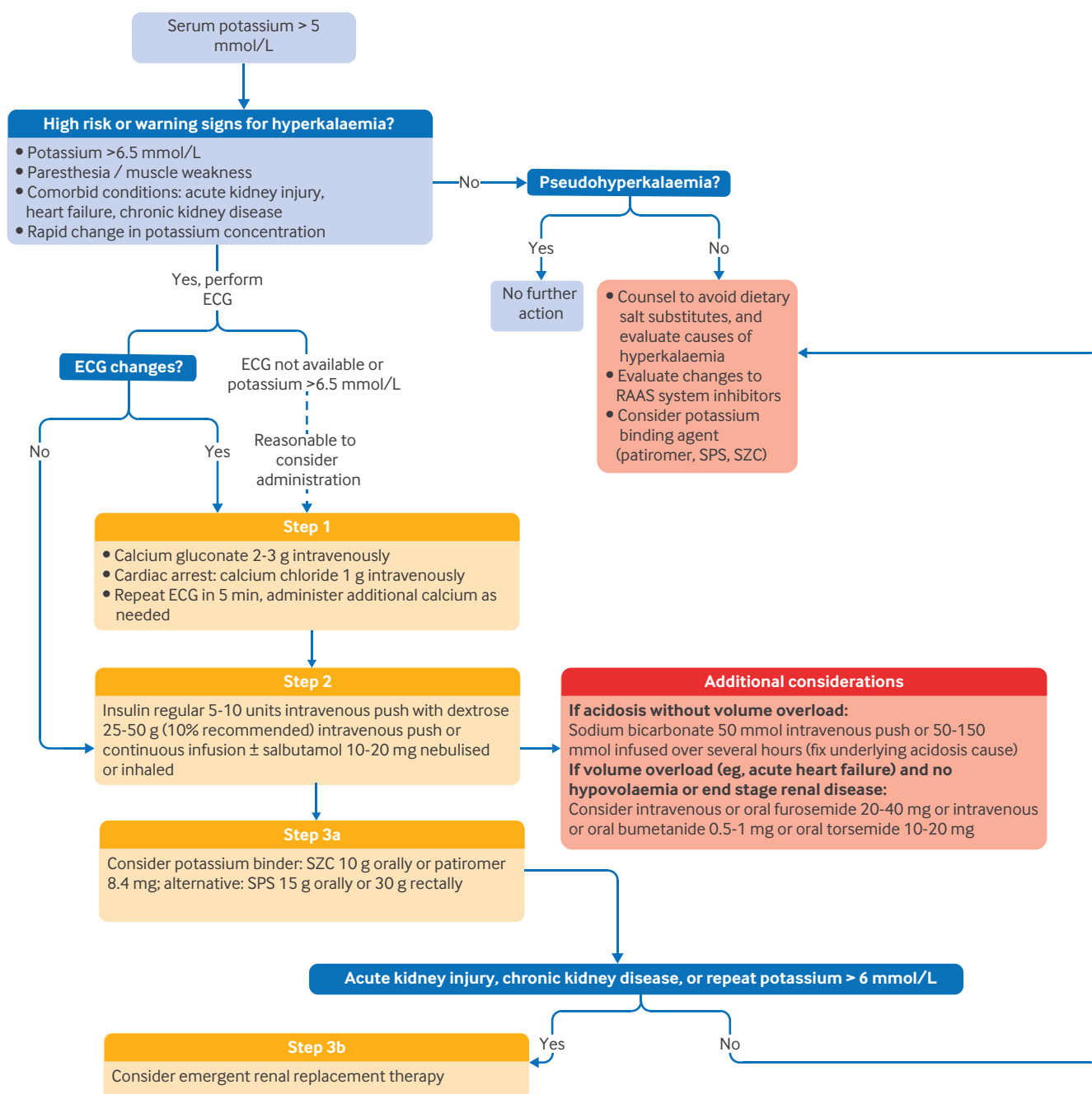


Fig 4 | Management algorithm for treating hyperkalaemia.^{1 5 6 59 77} ECG=electrocardiogram; RAAS=renin-angiotensin-aldosterone system; SPS=sodium polystyrene sulfonate; SZC=sodium zirconium cyclosilicate

a rapid estimate of potassium, or in prehospital or critical care transport; however, they often require laboratory confirmation. Whole blood analysers will not detect or signal haemolysis, which can mean a raised reading may be interpreted as abnormal when circulating potassium is normal. If a whole blood sample is abnormal, it should be confirmed with plasma potassium. Additionally, if a plasma sample is abnormal and detects haemolysis, a new sample should be obtained. A frequent clinical error is confirming an abnormal plasma sample with a whole blood sample, which will not indicate that haemolysis is present.⁷⁶

The most used laboratory tests for potassium measurement are the basic or comprehensive metabolic panel, which are performed on processed serum or plasma samples in the central laboratory. A key limitation is that specimen processing and transport can prolong turnaround times. A repeat serum or plasma sample should be obtained whenever haemolysis is reported to ensure accuracy of the level.^{67 75}

Management of hyperkalaemia

In patients with hyperkalaemia who are clinically unwell or have acute kidney injury, a three step management approach based on relative onset of action is recommended to prevent sequelae (fig 4).⁴ The first step is restoration of cardiac

conduction through calcium dependent channels. Guidelines recommend that this step should be undertaken rapidly if the patient has haemodynamic instability or is at high risk (eg, serum potassium >6.5 mmol/L).^{4 5 7 49 77} Step 1 is not necessary if hyperkalaemia is not an emergency or life threatening. Step 2 involves shifting potassium intracellularly and is prioritised after calcium administration. The third step entails enhancing potassium elimination from the body through pharmacotherapies (step 3a; eg, potassium binders) or renal replacement therapies (step 3b). The following section outlines treatments used during each step. However, hyperkalaemia management is complex and must be individualised according to the patient's clinical status (eg, presence of ECG changes, intravenous access, volume status, renal function, and acid-base balance) and institutional capabilities (eg, availability of emergent haemodialysis). Additionally, although numerous studies have evaluated hyperkalaemia treatments, interpretation is limited by heterogeneity in patient populations, clinical settings, dosing strategies, and timing of outcome assessment.^{15 42 78} Table 2 summarises all treatments across each phase.

Restoration of cardiac conduction

Restoration of cardiac conduction is achieved with the administration of intravenous calcium.⁷⁹ Historically it was thought that calcium led to membrane

Table 2 | Summary of drug treatments for hyperkalaemia^{5 15 77 99 101 113 140}

Drug	Initial dose	Pharmacokinetics	Expected reduction in serum potassium	Notes
Step 1: restoration of conduction				
Calcium gluconate 10%	1-2 g (10-20 mL) intravenously over 2-5 min or 3 g (30 mL) intravenously over 10 min	Onset 3 min; duration 30-60 min	None	Repeat dose in 5-10 min if ECG abnormalities do not normalise
Calcium chloride 10%	1 g (10 mL) intravenously over 2-5 min	Onset 3 min; duration 30-60 min	None	Central intravenous access preferred; use over gluconate in cardiac arrest
Step 2: intracellular shifting of potassium				
Regular insulin (with dextrose)	Insulin: 5-10 units or 0.1 unit/kg (maximum 10 units) intravenous push; dextrose: 25-50 g (10% recommended) intravenous push or continuous infusion	Onset <15 min; duration (insulin) 2 h; duration (dextrose) 1 h	0.6-1.4 mmol/L	Monitor glucose every 30-60 min for at least 4 h; administer subsequent doses of dextrose as needed; 5 units insulin recommended if CKD, acute kidney injury, insulin naive, weight <50 kg, or initial glucose <7 mmol/L (<126 mg/dL). Start at 25 g dextrose and repeat if needed. Consider starting dextrose infusion if patient is at high risk of hypoglycaemia (eg, end stage renal disease, acute kidney injury) or initial glucose <7 mmol/L (<126 mg/dL). Dextrose administration may not be needed if initial glucose >11.1 mmol/L (>200 mg/dL)
Salbutamol (albuterol)	10-20 mg nebulised or 4-6 puffs (metered dose inhaler)	Onset <30 min; duration 2 h	0.65-1 mmol/L	May be less effective in patients with end stage renal disease
Sodium bicarbonate	50 mmol intravenous push or 50-150 mmol infused over several hours	Variable	Negligible	Generally not recommended, can consider in select patients with metabolic acidosis
Step 3a: enhanced elimination				
Loop diuretics: furosemide, bumetanide	20-80 mg intravenously, 0.5-2 mg intravenously	Onset 30-60 min; duration 2-3 h	0.3 mmol/L	Higher listed doses may be required for patients with reduced renal function and those not naive to loop diuretics
Sodium polystyrene sulfonate	15 g orally, 30 g rectally	Onset 1 hour; duration 4-12 h	Variable and typically negligible	General use not recommended. Oral can be repeated up to four doses per day. Rectal dose can be repeated one additional time. Separate other oral drugs by 2-3 h to prevent decreased absorption
Patiromer	8.4-25.2 g orally	Onset 2-7 h	0.4-0.6 mmol/L	Separate other oral drugs by 2-3 h to prevent decreased absorption
Sodium zirconium cyclosilicate	10 g orally	Onset 1 h; duration 4-12 h	0.1-0.7 mmol/L	Can consider repeat doses for total of 30 g/day. Separate other oral drugs by 2-3 h to prevent decreased absorption

CKD=chronic kidney disease; ECG: electrocardiogram.

stabilisation by restoring the resting membrane potential; however, a recent canine myocyte study found that it restores conduction through calcium dependent propagation.⁸⁰ This finding has important implications for understanding calcium's role in hyperkalaemia, explaining its potential benefit in patients with signs of conduction slowing (ie, QRS widening) but not with T wave abnormalities.⁸¹ Consensus recommendations and guidelines state that calcium should be administered to patients with hyperkalaemia and ECG abnormalities.^{4 49 77} Calcium may be withheld in patients experiencing no ECG abnormalities and in mild hyperkalaemia (potassium <6 mmol/L).⁴ A lack of consensus exists on the use of calcium in patients with potassium >6 mmol/L without ECG abnormalities. The KDIGO conference summary of recommendations for the management of acute hyperkalaemia in the emergency department recommends calcium administration in all patients with potassium >6.5 mmol/L without immediate access to ECG.⁴⁹ In practice, a potassium concentration is rarely known before an ECG is accessible, but in that circumstance, it is reasonable to administer calcium.

There are two intravenous calcium salts routinely used: chloride and gluconate. Calcium chloride contains three times as much elemental calcium as calcium gluconate (for each 1 g: 6.8 mmol/10 mL or 273 mg elemental calcium v 2.3 mmol/10 mL or 93 mg elemental calcium, respectively).⁸² High quality data to guide calcium dosing are limited. The UK Kidney Association recommends a 3 g dose of calcium gluconate, while other organisations recommend 1-2 g or do not specify a dose.^{1 4 6 7} This guidance stems from a prospective, observational study of 111 patients who received calcium gluconate for hyperkalaemia with ECG changes, of whom 97% required three 1 g doses and a single 1 g dose was not effective for any patient.⁸¹ Because of its greater osmolality and irritant potential, calcium chloride can cause tissue necrosis in the setting of extravasation, and so should generally be reserved for central line administration or cardiac arrest. A recent retrospective study of 72 peripheral administrations of 10% calcium chloride showed an acceptable safety profile, with only 6% of the study population experiencing an infusion reaction, none of which caused permanent tissue injury.⁸³ Until higher quality data are available, calcium chloride is generally preferred through central access, though in critical situations it is appropriate to consider administration using large bore peripheral intravenous access. Other adverse effects seen with either preparation include hypotension, peripheral vasodilation, bradycardia, and arrhythmias, which appear to be dose and rate dependent.^{4 84} Additionally, it was originally thought that calcium gluconate required hepatic activation before exerting any clinical effect but this has since been disproved.⁸⁵ Despite these differences, both salts have been shown to have similar efficacy.^{85 86}

There is a paucity of high quality, prospective clinical trials comparing calcium salt formulations,

dosing strategies, or optimal timing of administration in hyperkalaemia management, therefore recommendations are based primarily on observational data and expert opinion.^{4 49 77 87} Despite this limitation, intravenous calcium is a cornerstone of treatment. In a prospective, multicentre observational study including 203 patients with hyperkalaemia, 55% received intravenous calcium.¹⁵ Additional evidence is needed to better elucidate calcium's ideal dose and role in special circumstances such as hyperkalaemia with isolated T wave abnormalities. Concerns also existed that calcium could precipitate ventricular arrhythmias in the presence of digoxin toxicity when hyperkalaemia is common. However, in a retrospective cohort study of 161 patients with digoxin toxicity, calcium administration was not associated with arrhythmias or increased mortality.⁸⁸ Lastly, although given initially to normalise the ECG, intravenous calcium does not have any impact on potassium concentration; after administration, the focus should be on shifting potassium intracellularly.

Summary

Step 1 for hyperkalaemia management includes rapid administration of intravenous calcium because it rapidly restores cardiac conduction and reduces arrhythmic risk, particularly in patients with evidence of conduction slowing such as QRS widening on the ECG. Current guidelines recommend intravenous calcium for patients with hyperkalaemia and ECG abnormalities, and empirical administration is reasonable when potassium >6.5 mmol/L and ECG acquisition is delayed; calcium gluconate is generally preferred for peripheral administration owing to a lower risk of tissue injury. Because intravenous calcium does not lower serum potassium, it should be followed promptly by treatments that shift potassium intracellularly and remove potassium from the body.

Intracellular shifting of potassium

The second phase in management is the intracellular shift of potassium. Treatments include insulin with dextrose, beta agonists, and sodium bicarbonate, which rapidly lower serum potassium by promoting cellular uptake, providing temporary reduction while definitive elimination strategies are started.^{5 49}

Insulin

Insulin lowers serum potassium by activating Na⁺/K⁺-ATPase, promoting an intracellular potassium shift within minutes and reducing serum potassium by approximately 0.6-1.4 mmol/L, lasting for several hours, though rebound hyperkalaemia may occur.^{89 90} Historically, 10-20 units of intravenous regular insulin was administered in combination with dextrose to prevent hypoglycaemia.⁴⁹ Emerging evidence from retrospective cohort studies suggests 5-10 units of regular intravenous insulin has a sufficient effect on potassium concentration with less hypoglycaemic effect, especially in patients with

renal disease.^{91 92} Several studies have compared 5 units or weight based dosing (0.1 unit/kg) to 10 units of regular intravenous insulin. A meta-analysis of 10 retrospective cohort studies (n=3437) showed that lower doses decreased odds of hypoglycaemia (odds ratio 0.55, 95% CI 0.43 to 0.69) with no difference in serum potassium reduction (mean difference -0.02 mmol/L, 95% CI -0.11 to 0.07) compared with 10 units.⁹³ However, two recent retrospective cohort studies reported conflicting findings for potassium lowering efficacy, highlighting the need for prospective studies to determine the optimal insulin dosing strategy.^{94 95} Rapid acting insulin (eg, lispro) has also been studied, with one retrospective trial of 237 patients showing a greater effect on serum potassium but higher risk of hypoglycaemia.⁹⁶ Similar pharmacokinetics would be expected with intravenous administration because distinctions between insulin formulations are largely based on subcutaneous absorption profiles.⁹⁷

Regardless of insulin formulation, hypoglycaemia remains the primary adverse effect that needs active mitigation. A scoping review including 62 studies found that hypoglycaemia occurred in 17.2% of patients in hospital receiving insulin for hyperkalaemia.⁹⁸ Risk factors included CKD, acute kidney injury, no history of diabetes, lower pretreatment glucose, lower body weight, receiving an insulin dose ≥ 10 units or dextrose doses ≤ 25 g.^{4 92 99 100}

Dextrose

Intravenous dextrose is typically administered with insulin to prevent hypoglycaemia, as a bolus or continuous infusion, with different concentrations used in practice (eg, 5%, 10%, 20%, or 50%). Lower concentrations (5%) can cause fluid overload, whereas hypertonic preparations (50%) may worsen hyperkalaemia, owing to osmotic effects shifting potassium from intracellular to the extracellular compartment, or cause extravasation injury.^{4 27} Additionally, close monitoring of blood glucose is necessary because the effect of dextrose is shorter than that of insulin (about one hour compared with two hours).¹⁰¹ A pretreatment blood glucose < 7 mmol/L (< 126 mg/dL) may signify high risk of iatrogenic hypoglycaemia.⁴ To prevent delays in dextrose administration, an as needed (ie, PRN) dextrose order can be built into hyperkalaemia order sets. Some studies have investigated a dextrose only approach to stimulate endogenous insulin release, especially in patients without diabetes.¹⁰² One small randomised crossover study found a modest reduction in potassium (0.50 ± 0.31 mmol/L) in 10 patients who received dextrose only that was numerically less compared with the reduction observed with insulin and dextrose (0.83 ± 0.53 mmol/L).¹⁰² However, available evidence in retrospective studies suggests a limited role for this approach because endogenous insulin release after dextrose administration is variable and may be blunted in patients with renal dysfunction, stress, or subclinical diabetes.

Beta agonists

Beta agonists like salbutamol (albuterol) also shift intracellular potassium through activation of Na^+/K^+ -ATPase and have been suggested as an additional or alternative treatment to insulin with dextrose.^{4 49} Salbutamol can be administered intravenously, nebulised, or through a metered dosed inhaler, all of which are considered equally efficacious.⁸⁷ Nebulised or inhaled salbutamol is preferred.⁷⁷ Salbutamol shows a dose dependent effect, with 20 mg doses leading to a larger reduction in serum potassium than 10 mg.^{4 5} Salbutamol has been shown to have an additive effect when used with insulin and dextrose. In a crossover study of 12 patients on haemodialysis, insulin-dextrose, salbutamol, or both decreased serum potassium by 0.65, 0.66, and 1.21 mmol/L, respectively, suggesting an additive effect.¹⁰³ An ongoing randomised controlled trial comparing these modalities is expected to be completed in 2026.¹⁰⁴ Adverse effects of beta agonists include tachycardia, tremors, and headache from beta receptor stimulation. Decreased effectiveness may be observed in patients with end stage kidney disease or those receiving non-selective beta blockers.^{4 103}

Sodium bicarbonate

Sodium bicarbonate has a modest and inconsistent effect on serum potassium, promoting intracellular potassium shift through increased sodium delivery and alkalisation, which enhance Na^+/K^+ -ATPase activity and H^+/K^+ exchange.⁷⁷ Adverse effects include hypernatraemia, alkalosis, and fluid overload. Intravenous coadministration of sodium bicarbonate and calcium should be avoided because of precipitation of calcium carbonate, which may occlude the intravenous line and impede delivery of drugs.¹⁰⁵ A second intravenous line can be placed or the line should be thoroughly flushed between administrations. An observational study of 203 patients with hyperkalaemia in the emergency department found that sodium bicarbonate was used in 29% within the first four hours of care.¹⁵ A healthy volunteer study in 14 patients showed that sodium bicarbonate is not effective in reducing serum potassium in patients without metabolic acidosis.¹⁰⁶ In patients with metabolic acidosis, the benefit of sodium bicarbonate is also questionable because the reduction appears to be independent of change in serum pH. Two prospective, randomised controlled trials evaluated the use of sodium bicarbonate in the management of metabolic acidosis; however, many of these patients did not have hyperkalaemia.^{107 108} Therefore, any potential benefit of sodium bicarbonate from these studies is likely from the treatment of the cause of metabolic acidosis and not the correction of hyperkalaemia. At present, evidence is lacking for sodium bicarbonate in the emergent treatment of hyperkalaemia in patients with or without metabolic acidosis.

Summary

Step 2 for hyperkalaemia management provides rapid but temporary reductions in serum potassium

while definitive elimination strategies are started. Insulin with dextrose remains the preferred first line treatment, with emerging evidence supporting lower insulin doses (5 units or 0.1 unit/kg) to reduce hypoglycaemia risk without compromising efficacy, though close glucose monitoring remains essential. Beta agonists such as nebulised salbutamol provide an additive potassium lowering effect and may be used adjunctively, whereas sodium bicarbonate has limited and inconsistent evidence for routine use except for select patients with marked metabolic acidosis.

Enhanced elimination of potassium

The third and last step in decreasing serum potassium concentration is enhancing the elimination. Loop diuretics, such as furosemide and bumetanide, inhibit the luminal sodium-potassium-chloride cotransporter in the loop of Henle.⁷⁷ Although effective for chronic hyperkalaemia management and prevention, loop diuretics are infrequently used in emergent treatment; one real world observational study reported use in only 5% of patients.¹⁵ Evidence supporting their role in acute hyperkalaemia is limited, and effectiveness depends on preserved kidney function and volume status, with risk of harm in patients with hypovolaemia or euvolaemia.^{7 49} In light of the available literature, this treatment should be reserved for patients with volume overload (eg, acute decompensated heart failure) and intact renal function.

Binding resins

Binding resins like sodium and calcium polystyrene sulfonate have been used for decades to treat hyperkalaemia by exchanging sodium or calcium for potassium in the gastrointestinal tract to enhance faecal potassium elimination.^{82 109} Despite historical use, evidence supporting their use in acute hyperkalaemia is limited.

A 2005 Cochrane systematic review on emergency interventions for hyperkalaemia references only two non-randomised studies from the 1960s showing a reduction in potassium concentration at 24 hours.^{109 110} A 2015 Cochrane systematic review including seven studies of 241 patients could not comment on sodium and calcium polystyrene sulfonate because there are no randomised controlled trials evaluating these agents.⁸⁷ Adverse effects include hypokalaemia, hypomagnesaemia, gastrointestinal upset, and rare but potentially fatal colonic necrosis.^{77 111} Additionally, these potassium binding resins carry the potential for drug interactions through impaired drug absorption.

Newer potassium binders, including patiromer and sodium zirconium cyclosilicate (SZC), have emerged with real world data from emergency departments suggesting potential benefits in patient disposition and hospital admission.¹¹² Patiromer is an oral agent that is approved for the chronic treatment of hyperkalaemia.¹¹³ A retrospective, single group inpatient cohort study (n=881) found a potassium

Table 3 | Summary of sodium zirconium cyclosilicate studies for treating emergent hyperkalaemia^{112 120-124 141-143}

Study first author, year	Study design	SZC intervention	Comparator	Results	Notes
Peacock, 2025	Experimental, randomised study, n=70	10 g up to three doses	Placebo	Absolute difference in potassium reduction at 4 h (mmol/L): -0.13, 95% CI -0.44 to 0.17	Significant reduction in potassium at 2 h (mmol/L): absolute difference -0.35, 95% CI -0.68 to -0.02
Grewal, 2025	Retrospective cohort study, n=101	5-10 g	SPS 15-30 g	No difference in primary outcome of median time to potassium normalisation (h): SZC 14 v SPS 17 (P=0.26)	Significant reduction in potassium for SZC group at 2 and 8 h; normokalaemia at 4 h for SZC compared with SPS
Thai, 2025	Retrospective cohort study, n=46	10-20 g (median)	SPS 30-45 g (median)	No difference in primary outcome of median time to potassium normalisation (h): SZC 26.7 v SPS 16.8 (P=0.24)	—
Joyce, 2024	Retrospective cohort study, n=246	5-15 g	SPS 15-30 g	No difference in decrease in mean±SD potassium at 24 h (mmol/L): SZC -0.78±0.81; SPS -0.91±0.76 (P=0.22)	—
Yoo, 2023	Retrospective cohort study, n=260	10 g	SPS 7.5-45 g	Non-significant reduction in mean±SD potassium (mmol/L): SZC -0.88±0.64; SPS -0.75±0.65 (P=0.09)	Potassium re-evaluated mean of about 15±8 h after administration
Hasara, 2023	Retrospective cohort study, n=105	10-20 g	SPS 15-60 g	No change in mean±SD repeat potassium between groups (mmol/L): SZC -1.1±0.87; SPS -1.1±0.62	Median time to repeat potassium (min): SZC 550; SPS 362
Gasparovic, 2025	Retrospective cohort study, n=189	Not specified but part of hyperkalaemic order set treatment	Control group receiving hyperkalaemic order set treatment	Non-significant reduction in mean±SD potassium at 6 h (mmol/L): SZC -1.096±0.71; non-SZC -1.067±0.6 (P=0.798)	—
Sullivan, 2023	Retrospective cohort study, n=3903	10 g (median)	SPS 30 g (median)	Significant reduction in mean potassium at 4-24 h in SPS group (mmol/L): SZC 0.78; SPS 0.96 (P<0.001)	Higher resolution rate of hyperkalaemia at 24 h in SPS group (74.9%) v SZC group (68.8%; P<0.001)
Bower, 2025	Retrospective cohort study, n=75	Single dose but amount not specified	SPS and patiromer: single dose but amount not specified	SZC and SPS had larger mean±SD reduction than patiromer at 24 h; SZC -0.92±0.85; SPS -1.16±0.74; patiromer -0.76±0.63 (P<0.001 for SZC and SPS)	—

CI=confidence interval; SD=standard deviation; SPS=sodium polystyrene sulfonate; SZC=sodium zirconium cyclosilicate.

reduction of 0.50 ± 0.56 mmol/L at six hours.¹¹⁴ An open label pilot randomised trial of 30 patients in the emergency department found that patiromer 25.2 g plus standard care reduced potassium within two hours compared with standard care alone, though no difference was observed at six hours.¹¹³ A larger study was started but subsequently stopped early after only enrolling 144 patients and the results have not been published outside of abstract form.^{115 116}

SZC is a non-absorbable inorganic cation exchanger that binds potassium and ammonium in exchange for sodium and hydrogen ions, with up to nine-fold greater potassium binding capacity than sodium polystyrene sulfonate.¹¹⁷ Adverse effects include gastrointestinal upset and oedema.⁸² In a recent case-control study of 500 patients, SZC use was associated with a reduced need for emergent haemodialysis compared with historical controls (odds ratio 0.23, 95% CI 0.17 to 0.31).¹¹⁸ However, a pilot randomised trial of 70 patients receiving SZC or placebo in addition to insulin and dextrose found no difference in potassium reduction at four hours, although SZC was associated with greater potassium reduction at two hours and less need for additional treatments at four hours.¹¹⁹ Several recent retrospective studies evaluating SZC for emergent hyperkalaemia have shown mixed results (table 3). These studies are limited in their retrospective design, heterogeneous dosing, concomitant pharmacotherapies for hyperkalaemia, and patient populations. Nonetheless, whether the hypothetical benefits of SZC are seen in a real world setting needs to be addressed.¹²⁰⁻¹²⁴ Unfortunately, a prospective study comparing sodium polystyrene sulfonate, SZC, patiromer, and placebo was terminated early with interim results showing no difference among the groups.^{125 126} Further prospective studies are needed to determine the efficacy of patiromer and SZC in treating emergent hyperkalaemia.

Haemodialysis

Haemodialysis is the most effective strategy for potassium elimination. Guidelines suggest it should be started in patients with hyperkalaemia refractory to medical treatment, end stage kidney disease requiring chronic haemodialysis, or severe acute kidney injury.^{4 5 7 77} Haemodialysis can reduce serum potassium by approximately 1 mmol/L within the first hour and up to 2 mmol/L within three hours, though rebound hyperkalaemia may occur 1-6 hours later, necessitating ongoing monitoring.¹⁰¹ Limitations include the time and logistics required to start haemodialysis, particularly in patients who are dialysis naive or when resources are limited, and adverse effects such as hypotension, cramping, headache, nausea, and vomiting.⁷⁷ Patients who are haemodynamically unstable may require the less efficient continuous renal replacement therapies. Another alternative if haemodialysis is not available or feasible is emergent peritoneal dialysis. In a recent prospective cohort study of 82 patients who were dialysis naive with end stage kidney disease and

hyperkalaemia, patients receiving peritoneal dialysis had similar reductions in potassium compared with haemodialysis on day 1 and similar stabilisation by days 5-15.¹²⁷ Temporising treatments such as intravenous calcium, insulin with dextrose, and salbutamol are often necessary while waiting for renal replacement therapy to start, particularly in patients with arrhythmias or haemodynamic instability.⁴

Summary

In summary, step 3 for hyperkalaemia management provides definitive treatment for hyperkalaemia, though evidence supporting pharmacological elimination strategies (step 3a) in the acute setting remains limited. Loop diuretics may be considered in patients with volume overload and preserved kidney function, while traditional potassium binders like sodium polystyrene sulfonate have delayed and inconsistent efficacy with important gastrointestinal adverse effects; SZC and patiromer show potential but require further prospective evaluation. Haemodialysis (step 3b) remains the most effective elimination strategy for severe or refractory hyperkalaemia, particularly in patients with renal failure, persistent ECG abnormalities, or inadequate response to temporising treatments.

Refractory hyperkalaemia

Hyperkalaemia refractory to standard treatment can be life threatening, necessitating urgent haemodialysis or continuous renal replacement therapy as a definitive treatment approach. Dialysis is the most effective method for potassium excretion, decreasing serum potassium by about 1 mmol/L in the first hour of treatment. Guidelines suggest that haemodialysis should be considered early in patients with advanced CKD, oliguric acute kidney injury (eg, <400 mL urine output in 24 hours), medically resistant hyperkalaemia, or hyperkalaemia resulting in cardiac arrest.^{77 82}

Special considerations

Drug induced hyperkalaemia

Drug induced hyperkalaemia presents major clinical challenges in managing outpatient and inpatient settings. Drugs can interfere with potassium homeostasis by promoting transcellular potassium shift, impairing renal potassium excretion, or through exogenous potassium intake, usually in the setting of acute kidney injury.¹²⁸ The reduction in renal potassium excretion owing to RAAS inhibition is the most common and important mechanism through which drugs are known to cause hyperkalaemia. A large observational study of 194 456 patients on antihypertensives reported an incidence of hyperkalaemia (defined as potassium >5 mmol/L) in 10.8% over three years, most commonly in those taking angiotensin converting enzyme inhibitors.¹²⁹ Despite this risk, KDIGO guidelines suggest not discontinuing RAAS inhibitors in patients with CKD unless they develop uncontrolled hyperkalaemia

Table 4 | Management of drug induced hyperkalaemia^{1 7 15 50 65 128 144-148}

Drug class	Mechanism of hyperkalaemia	High risk populations	Data	Management strategies
RAAS inhibitors (ACE inhibitors, ARBs)	Decrease in aldosterone—decrease in renal K ⁺ secretion	CKD, elderly, diabetes, heart failure, concomitant K ⁺ raising drugs	ACE inhibitors associated with higher odds of moderate hyperkalaemia v ARBs (odds ratio 1.26, 95% CI 1.02 to 1.56)	Do not routinely discontinue (KDIGO) in CKD unless uncontrolled hyperkalaemia; dietary K ⁺ restriction, stop K ⁺ supplements, add loop or thiazide diuretics or oral K ⁺ binders
Non-steroidal anti-inflammatory drugs	Inhibition of prostaglandins—decrease in renin and aldosterone	Older adults, CKD, diabetes, cardiovascular disease	16.7% developed hyperkalaemia or acute kidney disease, or both in older adults at high risk	Discontinue non-steroidal anti-inflammatories; enhance K ⁺ excretion with diuretics or binders if persistent
Calcineurin inhibitors (tacrolimus, ciclosporin)	Not fully elucidated; decrease in aldosterone activity, decrease in Na ⁺ delivery; Na ⁺ /Cl ⁻ cotransporter activation	Acute kidney injury, CKD	Incidence varies depending on risk (10-73%); higher with tacrolimus	Dose reduction if feasible; low K ⁺ diet, thiazide diuretics (eg, chlorthalidone), patiromer, or SZC; consider low dose fludrocortisone
Potassium sparing diuretics (spironolactone, eplerenone, amiloride, triamterene)	Aldosterone antagonism (spironolactone, eplerenone) or epithelial sodium channel blockade (amiloride, triamterene)	Elderly, heart failure, CKD, ACE inhibitor or ARB use	Increases odds of hyperkalaemia (2.1, 95% CI 1.3 to 3.3); increased hyperkalaemia associated hospital admissions after RALES study	Drug review, dose reduction or temporary discontinuation; long term K ⁺ binders and dietary counselling
Trimethoprim	Epithelial sodium channel blockade (amiloride-like effect)	CKD, high dose treatment	K ⁺ >5.5 mmol/L: about 10%; onset usually within one week	Discontinue drug; volume repletion; loop diuretics with or without intravenous fluids if continuation required
Succinylcholine (suxamethonium)	Depolarising NMBA—massive K ⁺ efflux	Burns, trauma, denervation, immobilisation, sepsis	Severe hyperkalaemia 5-15 days after traumatic or burn injury; risk persists for months	Avoid in patients at high risk; if cardiac arrest occurs: intravenous calcium, insulin, beta agonists; high dose insulin reported in case studies
Beta blockers (especially non-selective)	Decrease in cellular K ⁺ uptake through Na ⁺ /K ⁺ -ATPase; decrease in renin release	Diabetes, CKD, RAAS inhibitor use, cardiac bypass	Non-selective agents raise K ⁺ more than beta 1 selective; conflicting observational data	Weigh discontinuation carefully; dietary K ⁺ restriction, K ⁺ binders, dose reduction if necessary
Digoxin	Na ⁺ /K ⁺ -ATPase inhibition—K ⁺ efflux	CKD, acute kidney injury, hypothyroidism, drug interactions	Hyperkalaemia signals severe toxicity	Stop digoxin; digoxin-Fab for severe toxicity; intravenous calcium, insulin-glucose; haemodialysis if renal failure

ACE=angiotensin converting enzyme; ARB=angiotensin II receptor blocker; CKD=chronic kidney disease; CI=confidence interval; KDIGO: Kidney Disease: Improving Global Outcomes; RAAS=renin-angiotensin-aldosterone system.

despite medical treatment.⁷ Initial management should prioritise a stepwise approach including dietary potassium restriction, discontinuation of concurrent potassium supplements, and adding potassium wasting diuretics or oral potassium binders, or both to facilitate continued use of RAAS inhibitors. Table 4 lists common drugs that induce hyperkalaemia and potential management strategies.¹²⁸

Patients with CKD or end stage kidney disease on haemodialysis

In patients with advanced CKD, higher baseline potassium concentrations are tolerated because of chronic adaptive mechanisms in the distal nephron. Therefore, the 2024 KDIGO guidelines emphasise clinical context over absolute values.⁷ Moderate hyperkalaemia (potassium >6.0-6.4 mmol/L) should be managed in hospital in patients who are clinically unwell or with acute kidney injury, while severe hyperkalaemia (potassium ≥6.5 mmol/L) is a medical emergency requiring immediate confirmation of potassium levels and ECG monitoring.^{7 49} For patients with end stage kidney disease on haemodialysis, dialysis remains the mainstay of treatment for maintaining potassium homeostasis. However, managing chronic hyperkalaemia between haemodialysis sessions can be augmented by newer oral potassium binders (SZC, patiromer). These agents are especially important because they allow the continued use of RAAS inhibitors in patients with end stage kidney disease.¹³⁰

Cardiac arrest

In the setting of cardiac arrest with suspected hyperkalaemia, resuscitation efforts should align with the 2025 European Resuscitation Council and American Heart Association guidelines for cardiopulmonary resuscitation, which prioritise rapid stabilisation and intracellular potassium shift.^{1 6} These guidelines use indirect evidence from patients without cardiac arrest to suggest that intravenous insulin and dextrose provide potassium lowering effects in patients with cardiac arrest caused by suspected hyperkalaemia.^{1 6} However, clinicians must be vigilant in monitoring for hypoglycaemia during and after resuscitative efforts. Beta agonists beyond standard systemic adrenaline (epinephrine) provide no additional benefits in cardiac arrest. Sodium bicarbonate is indicated only for suspected hyperkalaemia related cardiac arrest caused by pre-existing metabolic acidosis.^{1 6} A recent randomised controlled trial of 391 patients with undifferentiated out-of-hospital cardiac arrest showed no benefit of calcium and signalled towards harm, and so it is not recommended.¹³¹ A suspected cause of hyperkalaemia was an exclusion criterion. A subgroup analysis showed similar results even in those with ECG changes suggesting hyperkalaemia. Overall, we need high quality clinical trials assessing the benefits of empirical calcium administration during cardiac arrest in patients with suspected hyperkalaemia.

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Table 5 | Summary of guideline recommendations for acute management of hyperkalaemia^{1 4-7 59 77 124}

Guideline (population of interest) and recommendations	Class of recommendation, level of evidence
Kidney Disease: Improving Global Outcomes 2024 clinical practice guideline for the evaluation and management of chronic kidney disease (patients with CKD)	
Consider decreasing dose or discontinuation of RAAS inhibitors in the setting of either symptomatic hypotension or uncontrolled hyperkalaemia despite medical treatment	Practice point
Variability in potassium laboratory measurements owing to various factors such as diurnal and seasonal variation, plasma versus serum samples, and drug impact	Practice point
Non-emergent pharmacological management of hyperkalaemia should be driven by local availability of agents and formulary restrictions	Practice point
Acute hyperkalaemia in the emergency department: a summary from Kidney Disease: Improving Global Outcomes Conference (acute management of hyperkalaemia)	
An ECG is recommended to assess the presence of hyperkalaemia induced ECG changes	Consensus suggestions
Rapid determination of potassium concentrations should be performed, preferably by point-of-care testing, followed by central laboratory confirmation	Consensus suggestions
Cardiac monitoring and 12 lead ECG for potassium concentrations >6.0 mmol/L	Consensus suggestions
Treatment—cellular membrane stabilisation:	Consensus suggestions
Intravenous calcium salts should be administered immediately in patients with hyperkalaemia presenting with ECG changes	
If potassium concentration is known but an ECG or placement on a monitor is not immediately possible, giving calcium to all patients with potassium concentration >6.5 mmol/L is suggested	
Treatment—potassium shift to the intracellular compartment:	
Short acting intravenous insulin (5-10 units) with glucose (25-50 g) administration is suggested for intracellular shift	
Beta agonists are suggested as an addition or alternative to insulin and glucose treatment	
Intravenous sodium bicarbonate is suggested to be used only in patients with metabolic acidaemia who are able to tolerate sodium load	
Treatment—potassium elimination:	
Consider loop diuretics unless patient has anuria or severe volume depletion	
Consider K ⁺ binders such as SPS (15-60 g orally) or SZC (10 g 3 times per day orally)	
Therapy refractory hyperkalaemia or hyperkalaemia with acute kidney injury should be referred to nephrology for consideration of haemodialysis	Consensus suggestions
Re-assessment should be performed 60 minutes after administration of potassium shifting agent and consider re-assessment at 2-3 hours after to assess for rebound hyperkalaemia	Consensus suggestions
2025 American Heart Association guidelines for cardiopulmonary resuscitation and emergency cardiovascular care (adults and paediatric patients in cardiac arrest)	
The effectiveness of intravenous calcium administration in adult and paediatric cardiac arrest owing to suspected hyperkalaemia is not well established	2b C-LD
The effectiveness of intravenous sodium bicarbonate administration in adult and paediatric cardiac arrest owing to suspected hyperkalaemia is not well established	2b C-EO
The effectiveness of intravenous insulin and glucose administration in adult and paediatric cardiac arrest owing to suspected hyperkalaemia is not well established	2b C-EO
Inhaled beta agonists for adults and paediatric patients in cardiac arrest owing to suspected hyperkalaemia is not recommended	No benefit C-EO
American Academy of Family Physicians: potassium disorders: hypokalaemia and hyperkalaemia (chronic and acute management and use of potassium binders)	
The decision for urgent hyperkalaemia treatment should not be based on ECG results alone because of lack of consistent threshold of electrocardiography changes	C
Consider reinitiating RAAS inhibitor treatment with potassium binders in patients with hyperkalaemia with CKD, heart failure, or diabetic nephropathy	C
UK Kidney Association 2023: clinical practice guidelines treatment of acute hyperkalaemia in adults (acute management of hyperkalaemia)	
All patients admitted to hospital with serum K ⁺ level ≥6.0 mmol/L have an urgent 12 lead ECG	1C
In emergencies, K ⁺ level is measured from an arterial or venous blood sample using point-of-care blood gas analyser while waiting for laboratory measurement	1B
Treatment of hyperkalaemia in hospital follows five step approach:	1B
Step 1: intravenous calcium administered in patients with hyperkalaemia in presence of ECG changes: 30 mL 10% calcium gluconate over 10 min or 10 mL 10% calcium chloride over 5 min (1C)	
Step 2:	
• Insulin-glucose (10 units insulin in 25 g glucose) by intravenous infusion is used to treat severe hyperkalaemia (K ⁺ ≥6.5 mmol/L) (1B)	
• Nebulised salbutamol 10-20 mg is used as adjuvant treatment for severe hyperkalaemia (K ⁺ ≥6.5 mmol/L) (1B)	
• Intravenous sodium bicarbonate infusion not used routinely for acute treatment of hyperkalaemia (2C)	
Step 3: SZC is recommended in emergency management of severe hyperkalaemia (K ⁺ ≥6.5 mmol/L; 1B) and suggested in acute management of moderate hyperkalaemia (K ⁺ 6.0-6.4 mmol/L; 1B)	
Step 4: serum K ⁺ assessment is recommended at least 1, 2, 4, 6, and 24 hours after identification and treatment of moderate or severe hyperkalaemia (2C)	
Step 5: prevent recurrence—no specific recommendations	
Patients having haemodialysis with severe hyperkalaemia (serum K ⁺ ≥6.5 mmol/L) recommended to receive dialysis treatment urgently	1A
Hyperkalaemia is considered in all patients who have a cardiac arrest as part of identifying and treating a reversible cause using the 4 Hs and 4 Ts approach	1A
Standard advance life support practice is recommended in cardiac arrest for patients requiring dialysis	1A
European Resuscitation Council guidelines 2025: executive summary (resuscitation in special circumstances in adults and paediatrics)	
Shift K ⁺ into cells: give 10 units soluble insulin and 25 g glucose intravenously to treat moderate to severe hyperkalaemia; follow with 10% glucose infusion at 50 mL/h for 5 h if pretreatment blood glucose <7 mmol/L	Expert consensus
Antagonise the effect of hyperkalaemia:	Expert consensus
• Use 30 mL 10% calcium gluconate intravenously for patients with severe hyperkalaemia with ECG changes	
• Administer 10 mL 10% calcium chloride intravenously and 50 mmol sodium bicarbonate in treating hyperkalaemic cardiac arrest	

(Continued)

Table 5 | Continued

Guideline (population of interest) and recommendations	Class of recommendation, level of evidence
Remove K ⁺ from the body	Expert consensus
• Give SZC 10 g orally	
• Consider dialysis for severe, refractory hyperkalaemia	
Consider extracorporeal cardiopulmonary resuscitation in accordance with local protocols if initial resuscitation attempt is unsuccessful	Expert consensus
Do not use calcium in children in cardiac arrest	Expert consensus
Hyperkalaemia management in the emergency department: an expert panel consensus (acute hyperkalaemia management in emergency department)	
Diagnosis: ECG should not be used as the sole method to rule out hyperkalaemia	—
Management algorithm:	—
• Rule out pseudohyperkalaemia by verifying potassium level before starting treatment, consider pretest probability of hyperkalaemia by reviewing medical history, current drugs, and laboratory abnormalities	
• Evaluate cardiac involvement with ECG. If new ECG changes are found, calcium gluconate 1 g is recommended and repeat as needed	
• Treatment options for hyperkalaemia are based on K ⁺ levels and categorised into redistribution (intravenous insulin and dextrose, nebulised beta 2 agonists) and elimination (oral binders, haemodialysis, diuretics). Intravenous sodium bicarbonate is not recommended for acute treatment of hyperkalaemia	
• Reassessment recommended every 2-4 h	
• Hospital admission for hyperkalaemia is recommended for patients in whom potassium only redistributed but not eliminated or hyperkalaemia is recalcitrant to acute treatment	
Practice point: consensus based statement representing expert judgment; consensus suggestions: based on review of current literature and conference discussions of acute hyperkalaemia. 2b=weak recommendation; C=consensus, disease oriented evidence, usual practice, expert opinion, or case series; 1=strong, 2=weak, A=high quality of evidence, B=moderate, C=low; C-EO=expert opinion; C-LD=limited data. CKD=chronic kidney disease; ECG=electrocardiogram; RAAS=renin-angiotensin system; SPS=sodium polystyrene sulfonate; SZC=sodium zirconium cyclosilicate.	

Pregnancy

Urgent management is indicated for severe hyperkalaemia or ECG changes. Pregnant patients with CKD are at risk of developing electrolyte imbalance and are at greater risk of having adverse maternal and fetal outcomes.¹³² Additionally, hypertensive disorders of pregnancy such as pre-eclampsia may increase risk of hyperkalaemia.¹³³ Pregnancy management is similar in acute settings to that for patients who are not pregnant by following a multimodal approach that focuses on transcellular shift and potassium excretion while identifying reversible causes.

Children

Hyperkalaemia in paediatric patients is defined as serum potassium concentration >5.5 mmol/L.¹ Common risk factors for hyperkalaemia are similar to those in adults, though less likely to be drug related. Although children with CKD can develop potassium abnormalities like adults, some may present with persistent hypokalaemia because of inherited or acquired renal tubular disorders.⁷ Managing hyperkalaemia in children requires an understanding of physiological differences and can be more nuanced. Owing to a lack of data from robust clinical trials, acute management strategies are frequently adapted from adult protocols while keeping drug dosing weight based.

Emerging treatments

New treatments in the development pipeline are limited. Cyclophosphamide analogues and polymer technologies are under early investigation for their potential to selectively bind potassium in the gastrointestinal tract with minimal adverse effects.¹³⁴ Emerging strategies for mitigating hyperkalaemia aim to improve potassium control while allowing

continuation of drugs that often precipitate hyperkalaemia (eg, RAAS inhibitors, potassium sparing diuretics).¹³⁵ Different RAAS inhibitors have different effects on potassium. For example, in the double blind randomised PARADIGM-HF (Angiotensin-Neprilysin Inhibition versus Enalapril in Heart Failure) trial of 8442 patients with class II-IV heart failure and reduced ejection fraction, the incidence of severe hyperkalaemia was lower with sacubitril-valsartan compared with enalapril (4.3% v 5.6%, P=0.007).¹³⁶ Similar results were found for sacubitril-valsartan and valsartan alone in the PARAGON-HF randomised trial including 4822 patients with heart failure and preserved ejection fraction.¹³⁷ Sodium-glucose cotransporter-2 (SGLT2) inhibitors may reduce hyperkalaemia risk while conferring cardiovascular and renal benefits.¹³⁸ A meta-analysis including 49 876 patients with type 2 diabetes at high cardiovascular risk or with CKD from randomised, double blind, placebo controlled clinical trials analysed the rate of serious hyperkalaemia (defined as serum potassium ≥6.0 mmol/L) in patients receiving SGLT2 inhibitors compared with placebo. SGLT2 inhibitors reduced the risk of serious hyperkalaemia (hazard ratio 0.84, 95% CI 0.76 to 0.93). Therefore, in patients with diabetes, SGLT2 inhibitors can decrease the risk of serious hyperkalaemia.¹³⁹ Combination regimens offer further promise for patients with complex or recurrent hyperkalaemia, reflecting a broader shift towards tailored, guideline supported strategies that maintain optimal disease modifying treatment without compromising safety.

Guidelines

Table 5 summarises and compares select key recommendations from major European and US guidelines and consensus statements on the acute

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QUESTIONS FOR FUTURE RESEARCH

- What is the safest and most effective insulin dosing strategy for acute hyperkalaemia that balances potassium reduction with minimisation of hypoglycaemia risk across diverse patient populations, including those with chronic kidney disease or end stage kidney disease?
- What is the comparative effectiveness of calcium gluconate versus calcium chloride in patients with hyperkalaemia and electrocardiogram abnormalities with respect to onset of myocardial stabilisation, duration of effect, and safety outcomes?
- Which patient and drug specific factors predict response or non-response to acute potassium shifting treatments (eg, insulin-dextrose, beta agonists), and how can these predictors inform personalised treatment algorithms?
- What role, if any, should newer potassium binders such as sodium zirconium cyclosilicate and patiomer have in the emergent treatment of hyperkalaemia, and can they meaningfully reduce the need for haemodialysis or hospital admission?
- What are the long term clinical and cardiovascular outcomes of maintaining RAAS (renin-angiotensin-aldosterone system) inhibitor treatment with adjunctive potassium lowering strategies compared with RAAS inhibitor discontinuation in patients with recurrent or chronic hyperkalaemia?

PATIENT INVOLVEMENT

The veteran engagement panel consisting of 11 veterans from three treatment facilities organised by the Edward Hines Jr VA Medical Centre reviewed the content of this paper.

Overall, patients emphasised the need for healthcare practitioners to communicate the risks of hyperkalaemia, diagnosis and treatment of hyperkalaemia to patients as follows:

- Use plain, patient centred language and reduce medical jargon when discussing hyperkalaemia with patients and caregivers.
- Clearly explain to those at risk of hyperkalaemia whether it develops gradually, occurs suddenly, or both.
- Clarify the association between hyperkalaemia, kidney disease, diabetes, heart disease, and arrhythmias in patients at risk.
- Discuss drug related potassium risk in practical terms, including why high risk drugs are prescribed and how patients are monitored.
- Provide clear guidance on symptoms and warning signs that require urgent or emergency evaluation.
- Encourage clinicians to create emergency action plans with patients and caregivers for worsening symptoms or emergency department visits.
- Include simple screening questions or symptom checklists to help identify patients at risk.
- Emphasise regular potassium monitoring and explain how patients are tracked over time.
- Promote direct clinician-patient communication rather than relying only on written educational materials.

diagnosis and management of hyperkalaemia. The sources were selected to provide guidance and a framework used by clinicians across different practice settings, including nephrology, emergency medicine, and critical care practitioners. While all guidelines aim to provide evidence based guidance, the strength and quality of evidence vary significantly.

Conclusion

Hyperkalaemia remains a common and potentially life threatening electrolyte disorder with important implications for patient safety across acute and chronic care settings. The burden of this condition is greatest among patients with CKD, heart failure, diabetes mellitus, and those receiving drugs that

impair renal potassium excretion, particularly RAAS inhibitors. Although the underlying mechanisms are well established, considerable variability persists in how hyperkalaemia is diagnosed, risk stratified, and managed in clinical practice. Emerging evidence, including improved understanding of potassium physiology and the introduction of newer potassium binding agents, has expanded therapeutic options. This review highlights the importance of individualised, context specific treatment strategies that balance rapid stabilisation with long term risk mitigation, particularly in high risk populations. Ongoing research is needed to standardise definitions, refine treatment algorithms, and better define the role of newer treatments to optimise outcomes for patients with hyperkalaemia.

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